

Jörg Weber

Curriculum Vitæ

Personal information

Title PD Dr. rer. nat.
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Work experience

01/2025– **Principal Investigator of FWF ESPRIT grant**, *University of Vienna*, Faculty of Mathematics
10/2023– **Moritz Schlick postdoctoral fellow**, *University of Vienna*, Faculty of Mathematics
12/2024
09/2023– **Junior fellow**, *Institut Mittag-Leffler*
10/2023
09/2022– **Researcher**, *Lund University*, Centre for Mathematical Sciences
08/2023
09/2020– **Postdoctoral fellow**, *Lund University*, Centre for Mathematical Sciences
08/2022
10/2016– **Research assistant**, *University of Bayreuth*, Department of Mathematics
08/2020

Education

05/2026 **Habilitation**, *University of Vienna*, Faculty of Mathematics, thesis: *Recent advances in the theory of steady water waves*
01/2017– **Dr. rer. nat.**, *University of Bayreuth*, Department of Mathematics, thesis: *The Relativistic Vlasov–Maxwell System with External Electromagnetic Fields*
07/2020
supervisor: Prof. Dr. Gerhard Rein
2014–2016 **Master**, *University of Bayreuth*, Mathematics, secondary subject Physics
2011–2014 **Bachelor**, *University of Bayreuth*, Mathematics, secondary subject Physics

Conferences and seminars

Organised events

- 2026 Co-organisation of minisymposium *Water Waves and Beyond* at *The 15th AIMS Conference* in Athens
- 2025– Co-organisation of *Partial Differential Equations and Mathematical Physics Seminar* at University of Vienna
- 2025 Co-organisation of minisymposium *Bifurcation and Pattern Formation in Dynamical Systems* at *Conference on Mathematics of Wave Phenomena* at Karlsruhe Institute of Technology
- 2024 Co-organisation of minisymposium *Steady Water Waves* at *SIAM Conference on Nonlinear Waves and Coherent Structures* in Baltimore
- 2023 Organisation of *Postdoc/PhD seminar* at Institut Mittag-Leffler, Stockholm
- 2021–2023 Co-organisation of *Seminar on Analysis, Geometry and PDEs* at Lund University

Talks and posters

- 2026 Talk at *SIAM Conference on Nonlinear Waves and Coherent Structures*, Montréal
Invited talk in working seminar, Hangzhou Normal University
Invited talk at workshop *Fluid Dynamics and Spectral Theory of Differential Operators*, Nanjing University
Invited short course (3 lectures) on ‘Travelling water waves,’ Nanjing University
Invited talk in *Séminaire Laurent Schwartz*, École Polytechnique
- 2025 Invited talk at *SIAM Conference on Analysis of Partial Differential Equations*, Pittsburgh
Invited talk at workshop *Interfaces, Instabilities, and Nonlocality in Fluid Dynamics*, University of Pittsburgh
Invited talk in *PDE and Analysis Seminar*, University of Pittsburgh
Invited talk at workshop *Modelling of fluid propagation: mathematical theory and numerical approximation*, International Centre for Mathematical meetings (CIEM), Castro Urdiales
Invited talk at workshop *Mathematical Advances in Geophysical Fluid Dynamics*, Oberwolfach Research Institute for Mathematics
Invited talk at *The Thirteenth International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory*, Athens, U.S.
Invited talk in *PDE Seminar*, Georgia Institute of Technology
Invited talk at *Conference on Mathematics of Wave Phenomena*, Karlsruhe Institute of Technology
- 2024 Invited talk at workshop *Nonlinear Water Waves: Rigorous Analysis and Scientific Computing*, Banff International Research Station
Invited talk at workshop *Mathematical theory of water waves*, Lund University
Invited talk in seminar *Partial differential equations*, University of Konstanz
Invited talk in seminar *Material modelling*, Weierstrass Institute for Applied Analysis and Stochastics, Berlin

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- Talk at *SIAM Conference on Nonlinear Waves and Coherent Structures*, Baltimore
- Invited talk at conference *Fluid Flows – Analysis and Modelling*, University of Vienna
- Invited talk at *GAMM Annual Meeting 2024*, University of Magdeburg
- Invited talk in *Applied Analysis Seminar*, Saarland University
- 2023 Invited talk in *MCMP Seminar* (for Master students in mathematics and physics), University of Vienna
- Invited talk in *Brown PDE Seminar*, Brown University
- Invited talk in programme *Order and Randomness in Partial Differential Equations*, Institut Mittag-Leffler, Stockholm
- Invited talk in programme *Mathematical Problems in Fluid Dynamics, part 2*, Simons Laufer Mathematical Sciences Institute, Berkeley
- Invited talk in *Differential Equations Seminar*, University of Missouri
- Invited talk in joint seminar *Asymptotic Models in Fluid Dynamics*, Lund University and University of Stuttgart
- 2022 Invited talk at *Vienna School of Mathematics*, University of Vienna
- Invited talk in *Analysis and Probability Seminar*, Chalmers University of Technology
- Poster in *Workshop on spatial dynamics and related approaches*, University of Stuttgart
- Talk at *SIAM Conference on Nonlinear Waves and Coherent Structures*, University of Bremen
- Invited talk in seminar *Nonlinear problems in Mathematical Physics*, University of Bayreuth
- Invited talk in workshop *New Directions in Water Waves*, University of Bath
- Poster at conference *When Kinetic Theory meets Fluid Mechanics*, ETH Zürich
- Invited talk in *DNA seminar*, Norwegian University of Science and Technology
- Poster at *SIAM Conference on Analysis of Partial Differential Equations* (held online)
- Talk at *Conference on Mathematics of Wave Phenomena*, Karlsruhe Institute of Technology
- 2021 Invited talk in *Differential Equations Seminar*, University of Missouri
- Talk at *4th IMA Conference on Nonlinearity and Coherent Structures*, Loughborough University
- Invited talk in *IntComSin Kolloquium*, University of Regensburg
- 2020 Talk in *Seminar on Analysis, Geometry, and PDEs*, Lund University
- Invited talk in *CAA seminar*, University of Erlangen-Nürnberg
- 2019 Talk at winter school *Gradient Flows and Variational Methods in PDEs*, University of Ulm
- Talk at conference *ENUMATH 2019*, Delft University of Technology
- Invited talk in *Analysis and Probability Seminar*, Chalmers University of Technology
- Talk at *GAMM Annual Meeting 2019*, University of Vienna
- 2017 Talk at *Young Researchers Meeting & CSE Workshop 2017*, University of Hamburg

Additional conferences and programmes participated in

- 2026 Programme *Mathematical Developments in Geophysical Fluid Dynamics*, Institut Henri Poincaré, Paris
Conference *Partial Differential Equations, Analysis and Geometry*, Institut des Hautes Études Scientifiques, Bures-sur-Yvette
- 2023 *Courant PIs Workshop* and annual meeting *Simons Collaboration on Wave Turbulence*, Courant Institute of Mathematical Sciences and Simons Foundation, New York
Abel symposium, Orkanger
Conference *Aspects of Nonlinear Evolution*, Leibniz University Hannover
- 2022 Seminar *Free Boundary Problems in Fluid Dynamics*, Oberwolfach Research Institute for Mathematics
Programme *Mathematical Perspectives of Gravitation beyond the Vacuum Regime*, Erwin Schrödinger Institute, Vienna (online)
- 2021 Programme *Mathematical Problems in Fluid Dynamics*, Mathematical Sciences Research Institute, Berkeley (online)

Teaching

Lectures

- 2024W *Advanced partial differential equations* (Master level), University of Vienna; designed the course content and created lecture notes; held all lectures and seminars (twice per week, in total 23 lectures, 7 seminars); held oral exams
- 2023S *Ordinary Differential Equations II* (Master level), Lund University; designed the course content and created lecture notes; held all lectures (twice per week, in total 18); designed and marked written and oral exams
- 2022S *Fourier Analysis* (Master level), Lund University; designed the course content and created lecture notes; held all lectures (twice per week, in total 16) and seminars (once per week, in total 8); designed and marked written and oral exams
- 2021W *Linear Analysis* (Bachelor level), Lund University; designed the course content and created lecture notes; held all lectures (twice per week, in total 18); designed and marked written and oral exams

Seminars

- 2025W *Fluid Dynamics* (Master level), University of Vienna; designed the course content, held lectures (once per week, in total 7) and created lecture notes in the first half of the course, followed by presentations by students in the second half
- 2025S *Analysis 2* (Bachelor level), University of Vienna; held exercise classes
- 2024S *Introduction to Mathematical Methodology* (Bachelor level), University of Vienna; held exercise classes
- 2024S *Bachelorseminar* (Bachelor level), University of Vienna; provided platform for students to work towards and present their Bachelor thesis
- 2020S *Advanced Complex Analysis* (Bachelor level), University of Bayreuth; held and organised seminars; marked written exams

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- 2019W *Introduction to Ordinary Differential Equations* (Bachelor level), University of Bayreuth; held and organised seminars; marked written exams
- 2018S *Introduction to Advanced Analysis* (Bachelor level), University of Bayreuth; held and organised seminars
- 2018W *Vector Calculus & Advanced Mathematics for Physicists B* (Bachelor level), University of Bayreuth; held and organised seminars; marked written exams
- 2014–2019 *Analysis 1* and *Analysis 2* (Bachelor level, several times), University of Bayreuth; held and organised seminars; (partly) designed and marked written exams

Supervision

Master's thesis

- 2023 Anna-Mariya Otsetova, *Axisymmetric capillary water waves on cylindrical fluid jets*, Lund University (co-supervised with E. Wahlén)

Bachelor's thesis

- 2026 Maris Rieck (B.Ed.), University of Vienna, ongoing
- 2026 Josef Hajda (B.Ed.), University of Vienna, ongoing
- 2025 Stefano Mazzeo, *Der Weierstraßsche Produktsatz und der Hadamardsche Produktsatz*, University of Vienna (co-supervised with L. Roberti)
- 2023 Abhijeet Vats, *The Path To Gelfand Duality & Spectral Theory*, Lund University (co-supervised with S. Pott)
- 2021 Yining Zhu, *An Elementary Proof of the Jordan Curve Theorem and its Application to the Poincaré–Bendixson Theorem*, Lund University (co-supervised with E. Wahlén)

Training

- 2025 Leadership summer school *LEAD_able*, LBG Career Center, Vienna
- 2025 Workshop *Diversity in Practice*, University of Vienna
- 2022S Course *Teaching and Learning in Higher Education*, Lund University

Grants

- 2026 Grant from *Fondation Mathématique Jacques Hadamard* for a research visit at École Polytechnique in Paris, 2,800 EUR
- 2025–2027 *ESPRIT* project *Overhanging water waves: new challenges and perspectives*, Austrian Science Fund (FWF), 340,819 EUR, DOI: [10.55776/ESP8360524](https://doi.org/10.55776/ESP8360524)
- 2025 *Research in Teams* programme *Waves interacting with the Antarctic Circumpolar Current*, Erwin Schrödinger Institute, Vienna, 4,800 EUR
- 2023 *The Fund of the Walter Gyllenberg Foundation* project *Localised pure-gravity water waves*, The Royal Physiographic Society of Lund, 46,000 SEK
- 2022 Travel funding for *Workshop on spatial dynamics and related approaches*, University of Stuttgart; *When Kinetic Theory meets Fluid Mechanics*, ETH Zürich; *Conference on Mathematics of Wave Phenomena*, Karlsruhe Institute of Technology (obsolete; conference was held online due to Covid-19), ca. 2,000 EUR

Fellowships and Awards

- 2023 *Junior fellowship*, Institut Mittag-Leffler, ca. 40,000 SEK
- 2023 *Seal of Excellence* of the European Commission for a high-quality project proposal submitted under the call for Marie Skłodowska-Curie Actions Postdoctoral Fellowships 2022
- 2017 Nominated for *Teaching Award of the Faculty of Mathematics, Physics & Computer Sciences*, University of Bayreuth

Service

Referee for *Adv. Nonlinear Anal.* / *Classical Quantum Gravity* / *Comm. Math. Phys.* / *Discrete Contin. Dyn. Syst. Ser. S* / *GEM Int. J. Geomath.* / *J. Differential Equations* / *J. Math. Fluid Mech.* / *Math. Methods Appl. Sci.* / *Math. Model. Anal.* / *Nonlinear Anal. Real World Appl.* / *Phys. D: Nonlinear Phenom.* / *Phys. Scr.* / *Plasma Phys. Control. Fusion* / *Proc. Appl. Math. Mech.* / *SIAM J. Control Optim.* / *Stud. Appl. Math.* / *Z. Angew. Math. Phys.*

- 2023 Examiner of Master's thesis of Francisco Carvalho, Lund University

Computer skills

C++, HTML, $\text{\LaTeX}$, Maple, Mathematica, MATLAB, Office

Languages

native German
fluent English
intermediate Swedish

Publications

- [1] (With A. Constantin). ‘On the propagation of mountain waves: Linear theory’. In: *Differential Integral Equations* 39.7–8 (2026), 457–495. DOI: 10.57262/die039-0708-457.
- [2] (With D. S. Seth, K. Varholm and E. Wahlén). ‘Rigidity of symmetric doubly-periodic water waves near shear flows’. In: *Nonlinearity* 39.3, 035015 (2026). DOI: 10.1088/1361-6544/ae49d8.
- [3] (With E. Lokharu). ‘An improved upper bound for the Froude number of irrotational solitary water waves’. In: *J. Fluid Mech.* 1030, A30 (2026). DOI: 10.1017/jfm.2026.11273.
- [4] (With A.-M. Otsetova and E. Wahlén). ‘Axisymmetric capillary water waves with vorticity and swirl connecting to static unduloid configurations’. In: *J. Differential Equations* 411 (2024), 604–618. DOI: 10.1016/j.jde.2024.08.005.
- [5] (With J. Bartsch, P. Knopf and S. Scheurer). ‘Controlling a Vlasov–Poisson plasma by a Particle-In-Cell method based on a Monte Carlo framework’. In: *SIAM J. Control Optim.* 62.4 (2024), 1977–2011. DOI: 10.1137/23M1563852.
- [6] (With E. Wahlén). ‘Large-amplitude steady gravity water waves with general vorticity and critical layers’. In: *Duke Math. J.* 173.11 (2024), 2197–2258. DOI: 10.1215/00127094-2023-0054.
- [7] (With E. Wahlén). ‘Global bifurcation of capillary-gravity water waves with overhanging profiles and arbitrary vorticity’. In: *Int. Math. Res. Not. IMRN* 2023.20 (2023), 17377–17410. DOI: 10.1093/imrn/rnac280.

- [8] (With E. Lokharu and E. Wahlén). ‘On the amplitude of steady water waves with favorable constant vorticity’. In: *J. Math. Fluid Mech.* 25.3, 58 (2023). DOI: 10.1007/s00021-023-00796-6.
- [9] (With A. H. Erhardt and E. Wahlén). ‘Bifurcation analysis for axisymmetric capillary water waves with vorticity and swirl’. In: *Stud. Appl. Math.* 149.4 (2022), 904–942. DOI: 10.1111/sapm.12525.
- [10] (With P. Knopf). ‘On the two and one-half dimensional Vlasov–Poisson system with an external magnetic field: global well-posedness and stability of confined steady states’. In: *Nonlinear Anal. Real World Appl.* 65, 103460 (2022). DOI: 10.1016/j.nonrwa.2021.103460.
- [11] ‘Weak solutions of the relativistic Vlasov–Maxwell system with external currents’. In: *Math. Methods Appl. Sci.* 44.6 (2021), 4770–4801. DOI: 10.1002/mma.7070.
- [12] ‘Optimal control of the two-dimensional Vlasov–Maxwell system’. In: *ESAIM Control Optim. Calc. Var.* 27, S19 (2021). DOI: 10.1051/cocv/2020069.
- [13] (With S. Günther, J. Körner, T. Lebeda, B. Pötzl, G. Rein and C. Straub). ‘A numerical stability analysis for the Einstein–Vlasov system’. In: *Classical Quantum Gravity* 38.3, 035003 (2021). DOI: 10.1088/1361-6382/abcdbf.
- [14] ‘Confined steady states of the relativistic Vlasov–Maxwell system in an infinitely long cylinder’. In: *Kinet. Relat. Models* 13.6 (2020), 1135–1161. DOI: 10.3934/krm.2020040.
- [15] (With P. Knopf). ‘Optimal control of a Vlasov–Poisson plasma by fixed magnetic field coils’. In: *Appl. Math. Optim.* 81.3 (2020), 961–988. DOI: 10.1007/s00245-018-9526-5.
- [16] ‘Hot plasma in a container—an optimal control problem’. In: *SIAM J. Math. Anal.* 52.3 (2020), 2895–2929. DOI: 10.1137/19M1275061.